

When Government spends, which sectors gain most?

Disaggregating the fiscal multiplier.

Michail Litainas

University of Sheffield

Abstract

This paper provides an empirical decomposition of the aggregate government spending multiplier into sectoral contributions. We estimate a Bayesian GVAR with a sectoral dimension that accounts for the heterogeneous allocation of government purchases across industries and the production-network structure of the economy. We exploit the model's granularity to trace the aggregate output response back to individual sectors. For the UK, we estimate an aggregate multiplier of around 0.55. Although Public Administration & Defence, Education, and Health receive more than 90% of government purchases and dominate the initial response, Trade, Manufacturing, Construction, and Health become the largest contributors at longer horizons. Counterfactual exercises further show that production-network linkages account for roughly 65% of the baseline multiplier. These results demonstrate that the industries receiving government demand are not necessarily those that ultimately account for the aggregate multiplier.

WORKING PAPER — COMMENTS WELCOME

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1 Introduction

Government spending is one of the most actively debated tools of macroeconomic stabilisation, yet its effects remain imprecisely understood. Governments allocate a substantial share of GDP to the direct purchase of goods and services, and these purchases are distributed across industries with markedly different production technologies, supply-chain positions, and labour intensities. Despite a large empirical literature quantifying the aggregate effects of fiscal policy (see [Ramey \(2019\)](#)), most studies treat government spending as a single aggregate and output as a homogeneous object, abstracting from their sectoral compositions. This abstraction is consequential. The aggregate fiscal multiplier is ultimately the outcome of heterogeneous responses of sectors and the interactions between them. Idiosyncratic government spending shocks need not remain concentrated in the industries receiving government demand: higher demand in directly exposed sectors can propagate to other parts of the economy through production linkages. Aggregate estimates therefore reveal little about which industries drive the fiscal response, how their contributions evolve as the shock propagates, or how these sectoral responses combine to produce the aggregate multiplier. This paper takes a step back from the aggregate and asks the following two questions: *Which sectors account for the aggregate government spending multiplier? How much of the aggregate multiplier is generated by cross-sectoral spillovers?*

A growing structural modelling literature has shown that sectoral heterogeneity and production networks can be important determinants of aggregate fiscal effects ([Cox et al., 2024](#); [Proebsting, 2022](#); [Flynn et al., 2022](#); [Bouakez et al., 2023](#); [Devereux et al., 2023](#); [Bouakez et al., 2025](#)). Empirical work has provided complementary evidence by exploiting granular spending data to estimate direct sectoral effects and the propagation of government demand through production networks. However, existing empirical approaches typically identify sector-specific spending shocks in panel settings and estimate responses that are pooled across industries ([Acemoglu et al., 2016](#); [Ramey, 2011](#); [Bouakez et al., 2023](#)). As a result, they are not designed to recover the heterogeneous contribution of each sector to the response to an aggregate fiscal shock. We therefore still know relatively little, empirically, about how the aggregate fiscal multiplier is built up from the responses of individual sectors.

This paper proposes a new empirical approach for decomposing the aggregate government spending multiplier into sectoral contributions. The methodology combines aggregate macroeconomic data, sectoral time-series, and Input-Output (IO) tables to estimate the sectoral effects of aggregate government expenditure and to trace the aggregate output response back to the individual industries that contribute to it. Our approach accounts for both sectors' direct exposure to government demand and the indirect effects that arise through intersectoral linkages. We estimate a Bayesian Global Vector Autoregression (GVAR) with a sectoral dimension that allows each industry to exhibit its own dynamics while interacting with the rest of the economy through the production network.¹ The sectoral allocation of the fiscal

¹Given the relatively short sample and the dimensionality of the sectoral system, we use Bayesian estima-

shock is disciplined by the historical distribution of government purchases, while intersectoral propagation is governed by the historical IO structure. Hence, in our setting, an aggregate spending shock represents an increase in government purchases that preserves their historically typical allocation across industries—a historically representative spending plan—rather than an isolated increase in purchases from a particular sector. The resulting framework provides a direct mapping from heterogeneous sectoral responses to the aggregate output response, allowing the aggregate fiscal multiplier to be decomposed into the contributions of individual sectors.

Our analysis focuses on the United Kingdom economy, using quarterly data covering 16 industries. We specify the deterministic weights of our GVAR using empirical regularities extracted from UK IO tables. We identify an unanticipated aggregate government expenditure shock while controlling for fiscal expectations, and trace its effects on sectoral gross value added (GVA), prices, and employment. Aggregating the sectoral responses provides a direct counterpart to the conventional aggregate fiscal response, allowing us to assess whether the multi-sector framework reproduces the dynamics obtained from a standard aggregate SVAR. We then use the sectoral output responses to decompose the aggregate government spending multiplier into industry-level contributions and conduct counterfactual exercises that isolate the role of intersectoral linkages in its propagation. To our knowledge, this is the first empirical framework to decompose the aggregate fiscal multiplier into sectoral contributions while simultaneously allowing for heterogeneous sectoral dynamics and propagation through production networks.

Our empirical framework consists of two interconnected blocks. The first comprises 16 sector-specific VARX models, which capture each industry’s own dynamics and its linkages to other industries through the input-output network. The second is a common-factor block containing aggregate macroeconomic variables, including government purchases, tax revenues, and interest rates, that affect all sectors. This structure gives rise to two channels of fiscal transmission. The direct channel captures each sector’s exposure to the aggregate fiscal shock, while the network channel captures the propagation of the shock across industries through intermediate-input trade. As directly exposed sectors expand production, they raise demand for inputs from their suppliers, generating an upstream demand-pull effect; at the same time, sectoral price responses alter the input costs faced by customer industries, transmitting cost pressures downstream.

Taken together, these network effects diffuse the fiscal impulse beyond the industries directly exposed to government demand, affecting both upstream suppliers and downstream users of intermediate goods. Since government purchases are distributed simultaneously across multiple sectors, the aggregate output response reflects both the direct effects of government demand and the subsequent amplification or attenuation of these effects through the production network. The framework therefore allows us to quantify how intersectoral

tion. The use of shrinkage priors helps regularise the parameter estimates and improves estimation precision relative to an unrestricted frequentist specification.

propagation contributes to the aggregate fiscal multiplier.

We obtain three main results. First, the multi-sector framework reproduces the aggregate dynamics of a conventional fiscal VAR while revealing substantial heterogeneity beneath the aggregate response. A 1% increase in government purchases raises aggregate output on impact, with the implied government spending multiplier increasing from around 0.35 initially to 0.55 after four quarters. The aggregate response obtained by weighting the sectoral GVA responses closely tracks that estimated from the benchmark aggregate SVAR, providing a validation of the sectoral framework. Beneath this aggregate response, however, sectoral contributions differ markedly in both magnitude and timing.

Second, the sectors that receive the largest direct increase in government demand are not necessarily those that account for the aggregate response at longer horizons. Public Administration and Defence, Education, and Health absorb more than 90% of government purchases and therefore dominate the initial response. As the shock propagates, however, the contribution of market sectors rises substantially. In particular, Trade, Manufacturing, and Construction become increasingly important contributors to aggregate output, reflecting the transmission of the fiscal impulse beyond the sectors directly exposed to government demand. The sectoral composition of the multiplier therefore changes over time as the shock diffuses through the production network.

Third, intersectoral linkages are quantitatively important for the size of the aggregate multiplier. Counterfactual exercises that shut down propagation through the input-output network reduce the impact multiplier from approximately 0.35 to 0.19, implying that network effects account for a substantial share of the aggregate output response. Both the upstream demand channel and the downstream cost channel contribute to this propagation, although their quantitative importance differs across sectors and horizons.

Our findings provide empirical support for a central implication of multi-sector models of fiscal policy: the aggregate effects of government spending depend not only on the sectors directly receiving government demand, but also on the propagation of the initial impulse through production networks. Our results complement structural studies showing that sectoral composition and input-output linkages shape fiscal multipliers by providing a purely empirical decomposition of these effects. They also complement empirical studies based on sector-specific spending shocks by showing how heterogeneous industry responses combine to generate the multiplier associated with an aggregate fiscal expansion.

Related Literature This paper contributes to several strands of the literature. It relates to the literature studying how sectoral heterogeneity and production networks shape aggregate fluctuations. A large theoretical literature shows how microeconomic shocks can generate aggregate effects through sectoral size and input-output linkages ([Gabaix, 2011](#); [Acemoglu et al., 2012](#); [Atalay, 2017](#)). More closely related, multi-sector models of fiscal policy show that the aggregate government spending multiplier depends on both the sectoral allocation of government purchases and the position of recipient industries in the production network

(Proebsting, 2022; Flynn et al., 2022; Bouakez et al., 2023; Devereux et al., 2023; Cox et al., 2024; Bouakez et al., 2025). Our analysis complements this structural literature by providing an empirical decomposition of these aggregate effects.

The paper is most closely related to the empirical literature on the sectoral effects of government spending. Nekarda and Ramey (2011) and Sabaj et al. (2023) document substantial heterogeneity in sectoral responses to government demand, while Acemoglu et al. (2016) and Barattieri et al. (2023) show that fiscal shocks propagate through input-output networks, particularly towards upstream suppliers. Related empirical exercises in Bouakez et al. (2023, 2025) further connect sectoral fiscal responses to industries' positions in the production network. These studies typically exploit sector-specific spending variation to estimate direct and network effects, often within panel settings. We instead start from an aggregate fiscal shock, allow individual industries to respond heterogeneously, and map these responses back into their contributions to the aggregate fiscal multiplier. This distinction allows us to quantify not only how fiscal shocks propagate across sectors, but also how much each sector and the production network contribute to the aggregate multiplier.

Roadmap: The remainder of the paper is organised in the following sections. [Section 2](#) discusses the structure of the Input-Output (IO) tables and extracts key empirical regularities from the data from the UK IO table that are used to specify our empirical model. [Section 3](#) discusses the econometric model, the data and the identification of the fiscal shock. [Section 4](#) discusses the results and [Section 5](#) concludes.

2 Stylised Facts from UK Input–Output Tables

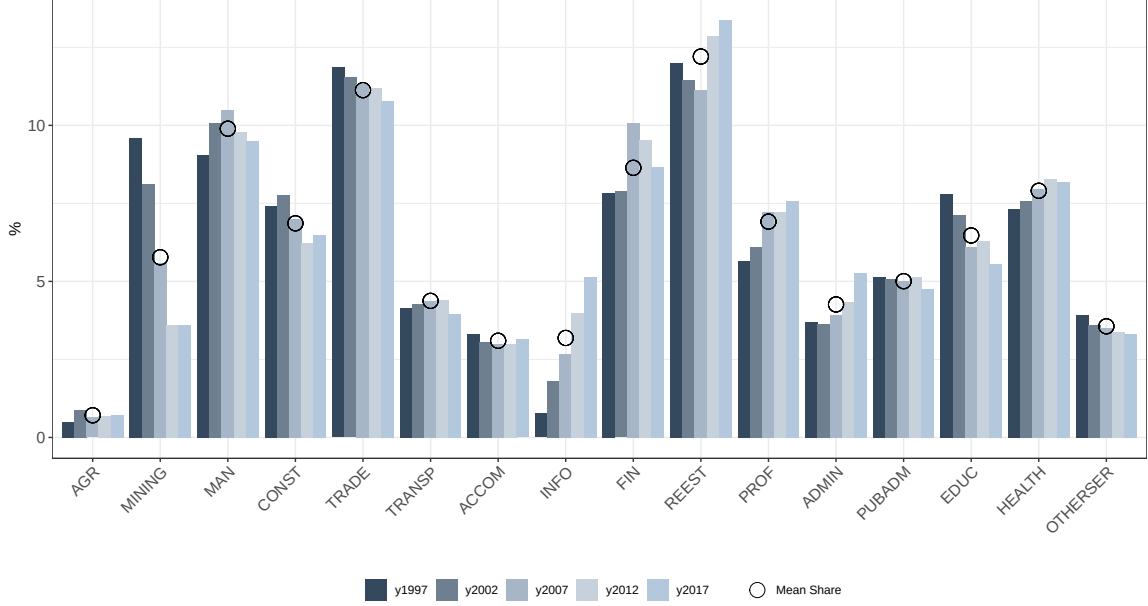
This section documents the empirical regularities used to specify our empirical model. We rely on UK IO tables to characterise three features of the economy over our sample: the relative size of industries, the sectoral allocation of government purchases, and the structure of intersectoral input linkages. These objects determine, respectively, the weights used to aggregate sectoral outcomes, the sectors' direct exposure to government demand, and the network structure through which shocks propagate across industries. We also examine the stability of these features over time, since substantial variation would weaken their interpretation as representative weights over the sample period.

The IO data are drawn from the OECD National Input–Output Table Dataset and cover the period 1997-2019. The original tables contain 45 industries, which we aggregate into 16 sectors to match the available quarterly series for sectoral GVA, employment, and prices. The 16 sectors are: (1) Agriculture, (2) Mining, Energy & Water, (3) Manufacturing, (4) Construction, (5) Trade, (6) Transportation, (7) Accommodation, (8) Information, (9) Financial, (10) Real Estate, (11) Professional Services, (12) Administration, (13) Public Administration & Defence, (14) Education, (15) Health, and (16) Other Services.²

²A complete description of the sector classification is provided in [Table A1](#).

Let s_i denote sector i 's share of aggregate output, m_{ij} the value of intermediate inputs supplied by sector i to customer sector j , and g_i final government demand for the output of sector i . These three objects underpin the weighting structure of the empirical model: s_i is used to aggregate sectoral outcomes, g_i determines sectors' exposure to government demand, and m_{ij} characterises the input-output linkages through which shocks propagate across industries.

Figure 1: Relative sector size over total production



Notes: The figure shows the sector size measured as a share of real sectoral GVA over real total output. Bars indicate relative size at five-year intervals from the start of the sample. Circles represent the average size of each sector over the entire 1997-2017 period. See Table A1 for a description of the sectoral aggregation and abbreviations.

Fact 1: Sectoral composition of output shows little change over time. We measure sectoral size by the share of real sectoral GVA in real GDP, defined as³⁴

$$share_{i,t} = \frac{GVA_{i,t}}{GDP_t}$$

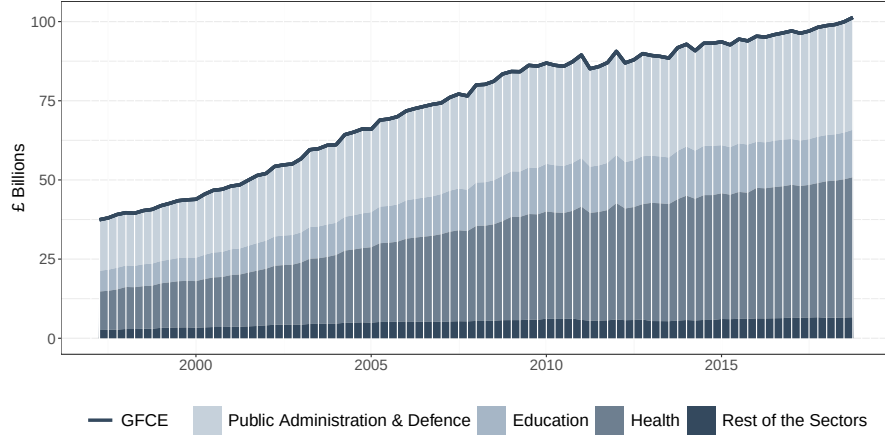
Figure 1 shows the sectoral output shares at five-year intervals from 1997 to 2017 (bars), while circles indicate the average share of each sector over the whole 1997-2019 sample. Industries maintain a fairly stable relative size throughout the sample period, albeit substantial differences in their relative size. The main exceptions are the Mining, Energy & Water sector, which exhibits a pronounced downward trend, and the Information sector, which share increased from less than 1% of GDP in 1997 to over 5% in 2017. Nevertheless, for most sectors the variation in the relative size is modest.

³⁴Data were deflated using the sectoral GVA price deflators from the ONS gdpolowlevelaggregates2022q4 dataset

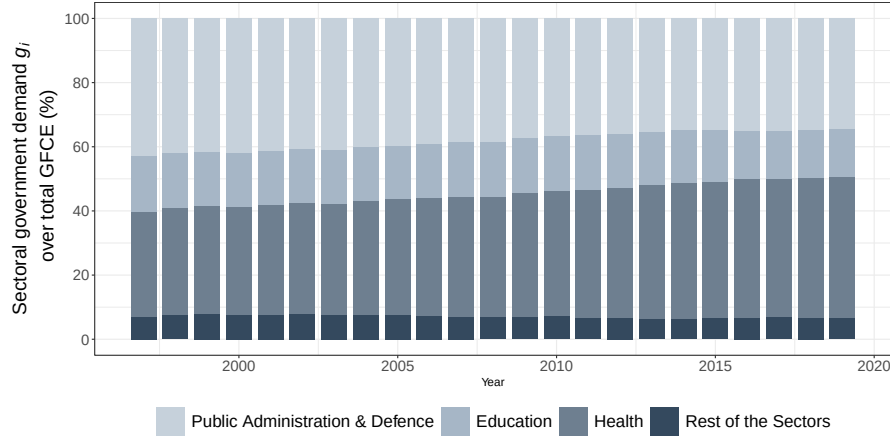
⁴Note that from national accounting $GDP = \sum_i GVA_i$

The relative stability of sectoral shares provides a useful benchmark for several aspects of the analysis. As discussed below, these shares play a central role in the specification of key model parameters, the aggregation of sectoral outcomes, and the measurement of each sector's contribution to the aggregate fiscal multiplier. Their stability ensures that the weighting scheme used throughout the analysis remains consistent over time, thereby facilitating a coherent mapping between sectoral dynamics and aggregate results.

Figure 2: Final Government Consumption Expenditure allocation across sectors



(a) UK GFCE breakdown by Sector



(b) Sectoral distribution of GFCE

Notes: The plot shows the total UK government spending (GFCE) from 1997 to 2019 and its allocation over the UK industries. Panel A shows data in absolute terms. Panel B shows the GFCE distribution in relative percentage terms.

Fact 2: Government spending is highly and systematically concentrated in three sectors. We measure each sector's exposure to government demand by its share of total government purchases,

$$gshare_{i,t} = \frac{g_{i,t}}{G_t}$$

where $g_{i,t}$ denotes government final demand for the output of sector i and $G_t = \sum_i g_{i,t}$ denotes aggregate government purchases.

Our measure of aggregate government purchases, G_t , is Government Final Consumption Expenditure (GFCE). GFCE records the total value of goods and services purchased by the general government, including compensation of government employees and purchases of goods and services from market producers. It includes both *government consumption*, referring to goods and services directly provided to the public, and *government gross investment*, which includes expenditures on fixed assets used in public service provision or government production. Importantly, unlike the broader notion of “government spending,” GFCE excludes direct monetary transfers, such as pensions, benefits, grants, and subsidies, while it includes social transfers in kind. This measure maps naturally into the government final-demand component of the IO tables, allowing aggregate government purchases to be written as $G_t = \sum_i g_{i,t}$, where $g_{i,t}$ denotes government final demand for the output of sector i .

Figure 2a shows the evolution of aggregate GFCE and its allocation across the 16 industries between 1997 and 2019. The solid line reports aggregate government purchases, while the stacked bars show the level of government demand directed towards individual sectors. Two features stand out. First, aggregate government purchases increase substantially over the sample, although their growth becomes more subdued after 2010. Second, this expansion is concentrated overwhelmingly in three sectors: Public Administration & Defence, Health, and Education. Government demand directed towards the remaining industries is small by comparison throughout the sample.

Figure 2b expresses the same expenditure in terms of sectoral shares and reveals the second important feature of government demand: its composition is remarkably stable over time. Public Administration & Defence and Health each account for approximately 38% of GFCE on average, while Education accounts for a further 16.6%. Together, these three sectors therefore receive more than 90% of total government purchases throughout most of the sample. The remaining industries collectively account for less than 10%, and no individual sector outside these three receives more than 2% of GFCE on average.⁵

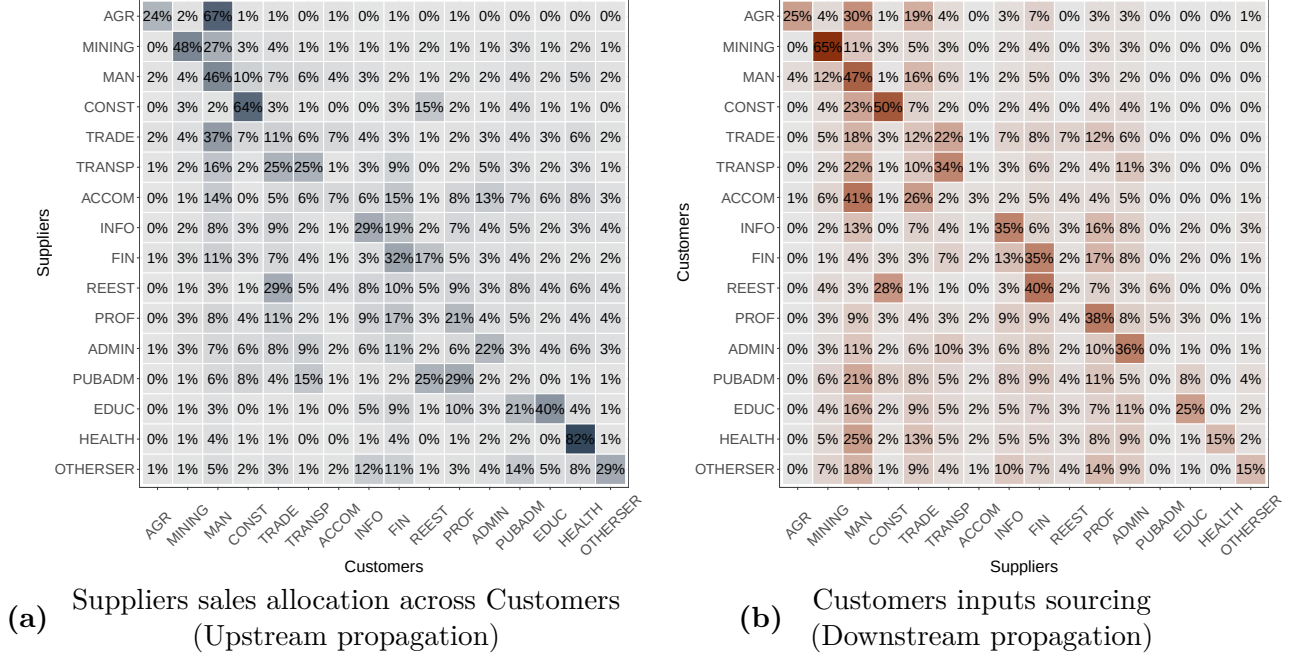
The combination of strong concentration and temporal stability is important for our empirical specification. Although the level of government purchases changes considerably over time, their relative allocation across industries changes much less. We therefore use the historical sectoral shares, $gshare_i$, to discipline sectors’ direct exposure to the aggregate fiscal shock. An increase in aggregate government purchases consequently preserves the historically representative sectoral composition of expenditure, rather than corresponding to an isolated increase in purchases from a particular industry.

Fact 3: Intersectoral IO linkages pattern has remained stable throughout our sample. The final set of weights characterises the production network through which shocks propagate across industries. Starting from the intermediate-input flows m_{ij} , where i denotes

⁵Table B3 reports summary statistics for government expenditure shares across all 16 sectors.

the supplier and j the customer sector, we construct two weighting matrices corresponding to the two directions of propagation in the model. For the upstream demand channel, we define

Figure 3: Input-Output Matrix Structures



Notes: The plot shows the production network linkages of the UK economy averaged over the 1997–2019 period. Panel (a) shows the row-normalised Supplier-Customer IO matrix. Panel (b) shows the row-normalised Customer-Supplier IO matrix.

$$w_{i,j}^{up} = \frac{m_{ij}}{\sum_j m_{ij}}$$

which measures the share of supplier i 's intermediate sales accounted for by customer j . These weights determine the extent to which changes in customer-sector activity affect the demand faced by their suppliers. For the downstream cost channel, we define

$$w_{i,j}^{down} = \frac{m_{ji}}{\sum_j m_{ji}}$$

which measures the share of customer i 's intermediate-input expenditure sourced from supplier j . These weights determine the extent to which changes in suppliers' prices affect the input costs faced by their customers.

Figure 3 shows the two matrices averaged over 1997–2019. Figure 3a reports the supplier-customer weights, w_{ij}^{up} , and therefore describes the direction of upstream demand propagation. Darker cells identify customers that account for a relatively large share of a supplier's intermediate sales. Figure 3b reports the customer-supplier weights, w_{ij}^{down} , which describe downstream cost propagation. Here, darker cells identify suppliers that represent a relatively important component of a customer's intermediate-input bundle.

The matrices reveal substantial heterogeneity in the structure of intersectoral linkages. Some industries sell intermediate inputs to a broad range of customers, while others are concentrated among a relatively small number of downstream sectors. Similarly, industries differ considerably in the composition of their intermediate-input bundles. Manufacturing, Trade, and Professional Services appear repeatedly as important nodes in the network, supplying inputs to a broad range of industries. The pronounced diagonal elements also indicate substantial within-sector intermediate use, reflecting transactions between firms belonging to the same broad industry.

Importantly for our empirical specification, the broad structure of these linkages changes relatively little over the sample period. [Figure B1](#) and [Figure B2](#) report the temporal variation of the supplier-customer and customer-supplier weights and show that, for most sector pairs, deviations around their historical averages are small. We therefore use the sample-average values to parameterise the GVAR weighting matrices. This provides a time-invariant representation of the production network while preserving the substantial cross-sectoral heterogeneity observed in the data.

3 Empirical framework

Our empirical analysis relies on the Global Vector Autoregressive (GVAR) framework of [Pesaran et al. \(2004\)](#), adapted to account explicitly for the heterogeneous allocation of government expenditures across sectors and the sectoral IO interconnections. The model consists of two interconnected blocks: (i) a system of 16 sector-level VARX models in which cross-sectoral interactions and common factors are explicitly modelled; (ii) a VARX of common factors that models the dynamics of the aggregate economy variables. Within this framework, government spending shocks—modelled as a common factor—affect each sector both directly, in proportion to its historical exposure to government demand, and indirectly, through the interdependencies arising from by input–output linkages. Because aggregate variables also respond to sectoral developments, the framework allows for two-way feedback between sectoral and aggregate dynamics.

[Figure 4](#) illustrates how fiscal shocks propagate through our model using a stylised three-sector economy — an upstream (US), midstream (MS), and downstream (DS) sector — linked by a common-factors block (CF) representing aggregate macroeconomic forces.

The fiscal shock propagates through the model through two channels. The first is direct: a government spending shock, identified in the common-factor block, affects each sector simultaneously according to its historical share in government purchases. Other aggregate variables in the common-factor block, such as tax revenues and interest rates, adjust endogenously and can transmit their effects to individual sectors. The second channel operates through the production network. When directly exposed sectors expand production, they raise their demand for intermediate inputs from supplier industries, transmitting the fiscal impulse upstream through a demand-pull effect. At the same time, sectoral price responses

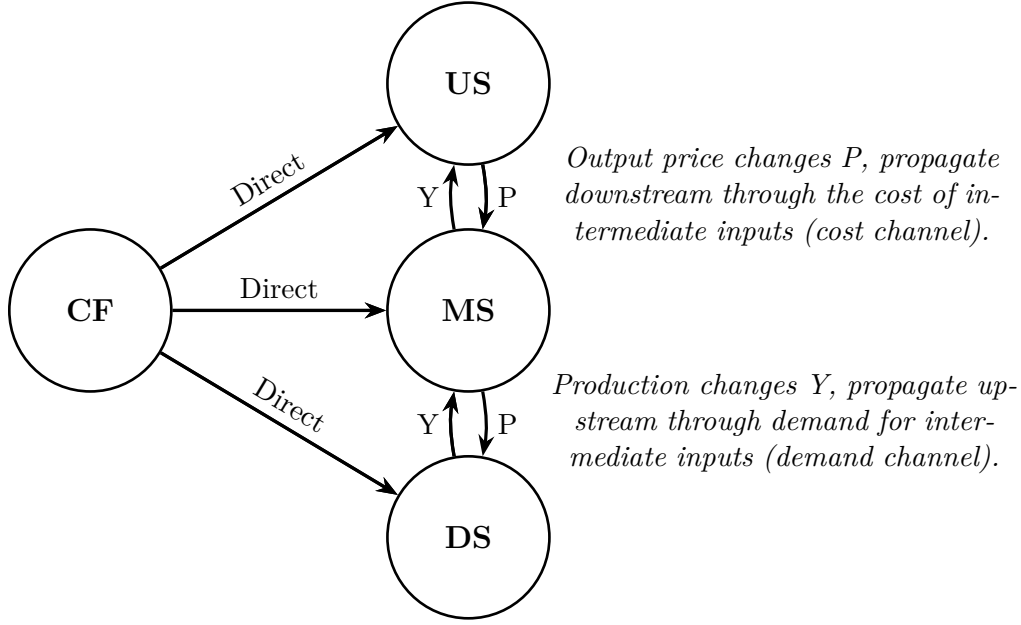


Figure 4: A simplified economy example showing the shock propagation channels captured by our model. CF stands for the common-factors block while US, MS and DS stand for upstream, midstream and downstream sectors respectively.

affect the cost of intermediate inputs faced by customer industries, generating a cost-push effect that propagates downstream.⁶

Taken together, the demand-pull and cost-push channels described above interact and diffuse the fiscal shock throughout the production network, affecting both upstream suppliers and downstream users of intermediate goods. Since government spending is typically distributed simultaneously across multiple sectors, the aggregate macroeconomic effect can be seen as the outcome of several idiosyncratic demand shocks that are amplified or dampened by the structure of the production network. Ultimately, quantifying these effects remains an important empirical question and it is the central objective of this paper.

Estimation is performed using bayesian shrinkage priors to mitigate over-parameterisation and improve efficiency given our short sample. The system is then solved in its global SVAR representation. Identification of the aggregate government spending shock is achieved via a standard Cholesky orthogonalisation, which, in this context, corresponds to an unanticipated contemporaneous 1% increase in government spending across all 16 sectors.

3.1 The Sectoral Models

Each sector is modelled as a VARX(p_i, q_i, q_i^x),

$$\mathbf{Y}_{it} = \sum_{l=1}^{p_i} \Phi_{i,l} \mathbf{Y}_{i,t-l} + \sum_{l=0}^{q_i} \Lambda_{i,l} \mathbf{Y}_{i,t-l}^* + \sum_{l=0}^{q_i^x} \Gamma_{i,l} \mathbf{S}_i \mathbf{X}_{t-l} + \mathbf{u}_{i,t} \quad (1)$$

⁶In the simplified linear production-chain example, a government spending shock originating in the most upstream or most downstream sector cannot propagate further in one direction because the network terminates at that sector. In the full model, by contrast, sectors are simultaneously connected to multiple suppliers and customers through a much richer production network.

where $\mathbf{Y}_{i,t}$ is a k_i -dimensional vector of endogenous sectoral variables. The k_i^* -dimensional vector $\mathbf{Y}_{i,t}^*$ contains the sector-specific *star* variables which represent developments in the other sectors of the economy and capture the relative spillovers. Vector $\mathbf{X}_t$ contains k_x common factors. The matrices $\Phi_{i,l}$, $\Lambda_{i,l}$, and $\Gamma_{i,l}$, with $l = 1, 2, \dots, p_i/q_i$, contain sector-specific coefficients. Hence, each industry is allowed to exhibit its own dynamics and to respond differently both to developments elsewhere in the production network and to aggregate macroeconomic conditions. Finally, $\mathbf{u}_{i,t}$ is a $k_i \times 1$ vector of sector-specific shocks that follows a white noise process with variance-covariance matrix $\Sigma_{u,i}$: $\mathbf{u}_{i,t} \sim iid(\mathbf{0}, \Sigma_{u,i})$.

Star variables are calculated as weighted averages of the corresponding sectoral variables of the rest $N - 1$ sectors. Weights are derived from IO tables as documented in [Section 2](#). Building on [Acemoglu et al. \(2016\)](#) and [Barattieri et al. \(2023\)](#), we distinguish between upstream and propagation. Unlike these studies though, that construct customer- and supplier-specific government spending shocks, we explicitly model the upstream propagation through sector customers' output, and the downstream propagation through sector suppliers' prices.

The intuition is straightforward. With our weighting scheme, each sector is affected by its:

- **Customers**, through changes in demand for intermediate inputs (upstream effect).
- **Suppliers**, through changes in the prices (cost) of intermediate inputs (downstream effect).

To illustrate, consider the case where sectoral model contains two endogenous variables: output (y_{it}) and prices (p_{it}). The corresponding “star” variables entering the model of sector i are defined as follows:

$$\mathbf{Y}_{i,t}^* = \begin{bmatrix} y_{i,t}^* & p_{i,t}^* \end{bmatrix}' = \begin{bmatrix} \sum_{j \neq i}^N w_{i,j}^{up} y_{j,t}, & \sum_{j \neq i}^N w_{i,j}^{down} p_{j,t} \end{bmatrix}', \quad (2)$$

where w_{ij}^{up} measures the share of sector i 's intermediate sales accounted for by customer j , while w_{ij}^{down} measures the share of sector i 's intermediate-input expenditure sourced from supplier j . Accordingly, $y_{i,t}^*$ captures changes in activity among sector i 's customers and therefore the demand-pull channel, while $p_{i,t}^*$ captures price developments among its suppliers and therefore the downstream cost channel.⁷

The vector $\mathbf{X}_t$ contains k_x the common factors, capturing the influence of economy-wide variables—such as government spending and monetary policy—on sectoral outcomes. These variables enter the sectoral models as weakly exogenous, while their dynamics are determined endogenously in the common-factor block described in the next section.

⁷We model cross-sectoral employment effects as propagating upstream and therefore construct the employment star variable using the same customer-based weights as for output. This assumption reflects the idea that stronger demand from downstream customers raises suppliers' production requirements and, consequently, their demand for labour.

Sector-specific heterogeneity in the transmission of common factors is captured by the diagonal weighting matrix

$$\mathbf{S}_i = \text{diag}(s_{i1}, s_{i2}, \dots, s_{ik_x}), \quad \mathbf{S}_i \in [0, 1],$$

where s_{ij} determines the exposure of sector i to the j -th common factor. Common factors whose effects are not scaled by sector-specific exposure, such as the interest rate, are assigned $s_{ij} = 1$, while setting $s_{ij} = 0$ excludes a factor from the sectoral model. Government purchases are treated differently because government demand is unevenly distributed across industries. For the government-spending common factor, we set $s_{i,G} = gshare_i$, where $gshare_i = \frac{1}{T} \sum_{t=1}^T \frac{g_{i,t}}{G_t}$ is sector i 's historical average share of total government purchases. Thus, the direct exposure of each sector to an aggregate government spending shock is proportional to the share of government demand it typically receives. The $gshare_i$ weights are derived from the Input–Output tables and documented in Section 2.

3.2 The Common-Factor Model

The common factors are modelled as a VARX(p_x, q_x),

$$\mathbf{X}_t = \sum_{l=1}^{p_x} \boldsymbol{\Phi}_{x,l} \mathbf{X}_{t-l} + \sum_{l=1}^{q_x} \boldsymbol{\Lambda}_{x,l} \mathbf{Y}_{x,t-l} + \mathbf{u}_{x,t}, \quad (3)$$

where $\mathbf{X}_t$ is a k_x -dimensional vector of aggregate macroeconomic variables and $\mathbf{Y}_{x,t}$ is a k_x^* -dimensional vector capturing feedback from sectoral developments to the aggregate economy. The coefficient matrices $\boldsymbol{\Phi}_{x,l}$ and $\boldsymbol{\Lambda}_{x,l}$ govern, respectively, the dynamics of the common factors and their response to lagged sectoral developments. Finally, $\mathbf{u}_{x,t}$ is a $k_x \times 1$ vector of aggregate innovations, with $\mathbf{u}_{x,t} \sim iid(\mathbf{0}, \boldsymbol{\Sigma}_{u,x})$.

The feedback variables are constructed by aggregating the corresponding sectoral variables,

$$\mathbf{Y}_{x,t} = \sum_{i=1}^N \mathbf{W}_i \mathbf{Y}_{i,t}, \quad (4)$$

where $\mathbf{W}_i$ contains the aggregation weights appropriate for each sectoral variable. For output, the weights correspond to each industry's share of aggregate GVA, so that sectoral output developments map directly into aggregate output. Employment is aggregated using the corresponding sectoral employment shares, while sectoral prices are aggregated using sectoral output weights to obtain an aggregate price measure. The use of the same aggregation structure throughout the model ensures a consistent mapping between sectoral and aggregate dynamics.

Sectoral feedback enters the common-factor block only with a lag. This restriction rules out contemporaneous feedback from sectoral variables to aggregate variables within the individual common-factor VARX and avoids simultaneity between the two blocks. Once the

sectoral and common-factor models are combined in the global representation, however, aggregate and sectoral variables are jointly endogenous and interact dynamically.

3.3 The Global Solution

The sectoral VARX models and the common-factor block are estimated separately and then combined into a single global system. Let

$$\mathbf{Z}_{x,t} = \begin{bmatrix} \mathbf{X}_t \\ \mathbf{Y}_{x,t}^* \end{bmatrix}, \quad \mathbf{Z}_{i,t} = \begin{bmatrix} \mathbf{Y}_{i,t} \\ \mathbf{Y}_{i,t}^* \\ \mathbf{S}_i \mathbf{X}_t \end{bmatrix},$$

denote the stacked vectors entering the common-factor and sectoral models, respectively. Each block can be written compactly as

$$\mathbf{A}_{s,0} \mathbf{Z}_{s,t} = \sum_{\ell=1}^p \mathbf{A}_{s,\ell} \mathbf{Z}_{s,t-\ell} + \mathbf{u}_{s,t}, \quad s \in \{x, 1, \dots, N\}, \quad (5)$$

where $p = \max_s \{p_s, q_s\}$, with coefficient matrices set equal to zero whenever the corresponding lag is not included in a particular block.

Define the global vector of endogenous variables as

$$\mathbf{Y}_t = [\mathbf{X}_t' \quad \mathbf{Y}_{1,t}' \quad \dots \quad \mathbf{Y}_{N,t}']'.$$

Because both the sector-specific star variables and the aggregate feedback variables are weighted combinations of elements of $\mathbf{Y}_t$, each block-specific vector can be written as

$$\mathbf{Z}_{s,t} = \mathbf{W}_s' \mathbf{Y}_t, \quad (6)$$

where $\mathbf{W}_s$ is a block-specific linking matrix containing the IO weights, sectoral aggregation weights, and common-factor exposure weights described above.

Substituting (6) into (5) and stacking the common-factor and sectoral equations yields

$$\mathbf{G}_0 \mathbf{Y}_t = \sum_{\ell=1}^p \mathbf{G}_\ell \mathbf{Y}_{t-\ell} + \mathbf{u}_t, \quad (7)$$

where $\mathbf{G}_0$ collects the contemporaneous relationships across blocks, $\mathbf{G}_\ell$ collects the corresponding lagged coefficients, and $\mathbf{u}_t$ stacks the innovations from the common-factor and sectoral models.

Provided that $\mathbf{G}_0$ is nonsingular, the global system can be written in reduced form as

$$\mathbf{Y}_t = \sum_{\ell=1}^p \mathbf{F}_\ell \mathbf{Y}_{t-\ell} + \tilde{\mathbf{u}}_t, \quad (8)$$

where

$$\mathbf{F}_\ell = \mathbf{G}_0^{-1} \mathbf{G}_\ell, \quad \tilde{\mathbf{u}}_t = \mathbf{G}_0^{-1} \mathbf{u}_t.$$

The global representation therefore endogenises the variables that enter the individual VARX blocks as external or weakly exogenous. It captures the contemporaneous and dynamic interactions between the aggregate economy and the 16 sectors, and forms the basis for the identification of the fiscal shock and the computation of impulse responses.

3.4 Model specification and identification

Specification: We specify a seven-variable VARX model for the common-factors block as follows:

$$\mathbf{X}_t = \begin{bmatrix} G_t & \Delta G_t^{fc} & T_t & R_t^S & R_t^L & S_t & V_t \end{bmatrix}', \quad \mathbf{Y}_{x,t-1}^* = \begin{bmatrix} Y_t^* & N_t^* & P_t^* \end{bmatrix}' \quad (9)$$

where G_t is the real government purchases (GFCE), ΔG_t^{fc} is the forecast of government spending growth, T_t is the real tax revenues, R_t^S and R_t^L are the short and long-term interest rates, S_t is the real effective exchange rate and V_t is the real stock market price index. The feedback effects vector $\mathbf{Y}_{x,t-1}^*$ contains three aggregated sectoral variables which are equivalent to aggregate output Y_t^* , employment N_t^* and GDP deflator P_t^* .⁸

Following [Born et al. \(2013\)](#); [Auerbach and Gorodnichenko \(2013\)](#) and [Ilori et al. \(2022\)](#) among other, we include cumulative projections of government spending growth, ΔG_t^{fc} , to account for fiscal foresight. Government expenditure decisions are often announced before their implementation, so innovations in realised spending may contain information that was already anticipated by private agents. Including government spending forecasts therefore helps distinguish unexpected changes in government purchases from previously anticipated fiscal developments. We construct ΔG_t^{fc} using historical vintages of the OECD Economic Outlook. Since the forecasts are published biannually, we interpolate the series to quarterly frequency using midpoint averaging.

We further augment the common-factor block with forward-looking financial variables, including the long-term interest rate, the real exchange rate, and stock prices. Following [Ellahie and Ricco \(2017\)](#), these variables broaden the information set available to the econometrician and help mitigate informational insufficiency in the identification of fiscal shocks.

We specify the sectoral vectors as:

$$\mathbf{Y}_{i,t} = \begin{bmatrix} Y_{i,t} & N_{i,t} & P_{i,t} \end{bmatrix}', \quad \mathbf{Y}_{i,t}^* = \begin{bmatrix} Y_{i,t}^* & N_{i,t}^* & P_{i,t}^* \end{bmatrix}' \quad (10)$$

where $Y_{i,t}$ is the real sectoral GVA, $N_{i,t}$ is the sectoral employment and $P_{i,t}$ is the sectoral GVA deflator. The corresponding star variables capture developments in sectors connected to industry i through the production network, as described in [Section 3.1](#).

⁸The common factors block resembles a typical aggregate economy VAR, where fiscal and other variables are ordered first before output, prices and employment, but with the last three equations of the VAR system being redundant.

Sectoral GVA, employment, and price data are obtained from the ONS, while the aggregate macroeconomic variables and fiscal forecasts are obtained from the OECD. Sectoral deflators are constructed as the ratio of nominal to real GVA, and the series are seasonally adjusted where necessary. [Table B2](#) provides detailed definitions and sources.

The model is estimated over the 1997Q1-2019Q4 sample with a constant and a trend, 2 lags for the endogenous and 1 lag for the star variables ($p_i = p_x = 2, q_i = q_x = 1$). For the estimation of the individual VARX* models we use a standard Normal Inverse-Wishart prior in Gibbs sampling algorithm. We kept the last 15000 posterior draws of 30000 draws in total.

Identification of the Government Spending Shock We are interested in identifying an aggregate, unanticipated government spending shock originating in the common-factor block. Let the reduced-form innovations of the global model satisfy

$$\tilde{\mathbf{u}}_t = \mathbf{P}\boldsymbol{\varepsilon}_t, \text{ with } E(\boldsymbol{\varepsilon}_t\boldsymbol{\varepsilon}_t') = \mathbf{I}, \quad (11)$$

where $\boldsymbol{\varepsilon}_t$ denotes the vector of orthogonal structural innovations and $\mathbf{P}$ maps structural shocks into the reduced-form residuals as follows:

$$\mathbf{P} = \begin{bmatrix} \mathbf{P}_{x,x} & \mathbf{P}_{x,1} & \dots & \mathbf{P}_{x,N} \\ \mathbf{P}_{1,x} & \mathbf{P}_{1,1} & \dots & \mathbf{P}_{1,N} \\ \dots & \dots & \dots & \dots \\ \mathbf{P}_{N,x} & \mathbf{P}_{N,1} & \dots & \mathbf{P}_{N,N} \end{bmatrix} \quad (12)$$

Because our interest lies exclusively in the aggregate government spending shock, structural restrictions are imposed on the common-factor block. We identify the shock recursively, following the standard fiscal VAR approach of [Blanchard and Perotti \(2002\)](#). Government purchases are ordered first in the common-factor vector, so that government purchases do not respond contemporaneously to innovations in the remaining macroeconomic variables, whereas those variables may respond within the quarter to an innovation in government purchases. The corresponding structural impact matrix for the common-factor block, $\mathbf{P}_{xx}$, is obtained from the Cholesky factorisation of its reduced-form covariance matrix.

The identified innovation is an aggregate government spending shock. Its direct effect on individual industries is governed by the sector-specific exposure matrix $\mathbf{S}_i$. As described above, the government-spending element of this matrix is

$$s_{i,G} = gshare_i,$$

where $gshare_i$ is sector i 's historical average share of total government purchases. Consequently, the fiscal experiment does not correspond to an independently identified spending shock in a particular industry. Instead, it represents an unexpected increase in aggregate government purchases whose direct sectoral incidence preserves the historical composition of government demand. We normalise the shock so that aggregate government purchases increase by 1% on impact.

We do not seek to identify the sector-specific innovations. Accordingly, we normalise the sector-specific diagonal blocks of $\mathbf{P}$ to identity matrices and set all off-diagonal blocks to zero. Under these restrictions, the structural impact matrix takes the block-diagonal form $\mathbf{P} = \text{diag}[\text{chol}(\boldsymbol{\Sigma}_{x,x}), \mathbf{I}_{k_1}, \dots, \mathbf{I}_{k_N}]$.

Setting the off-diagonal blocks of $\mathbf{P}$ to zero does not prevent the aggregate fiscal shock from affecting individual sectors contemporaneously. Rather, $\mathbf{P}$ identifies the structural innovations, while their transmission across blocks is governed by the structure of the GVAR. In particular, the government spending shock reaches sector i directly through the common-factor exposure matrix $\mathbf{S}_i$, and propagates across industries through the contemporaneous input-output linkages embedded in $\mathbf{G}_0^{-1}$. Impulse responses to the government spending shock are therefore computed from the global representation as

$$\Delta \mathbf{Y}(h) = \mathbf{F}^h \mathbf{G}_0^{-1} \mathbf{P} \mathbf{d}_G,$$

where $\mathbf{d}_G$ is a selection vector whose element corresponding to the government spending innovation equals one and all remaining elements equal zero.

4 Empirical Findings

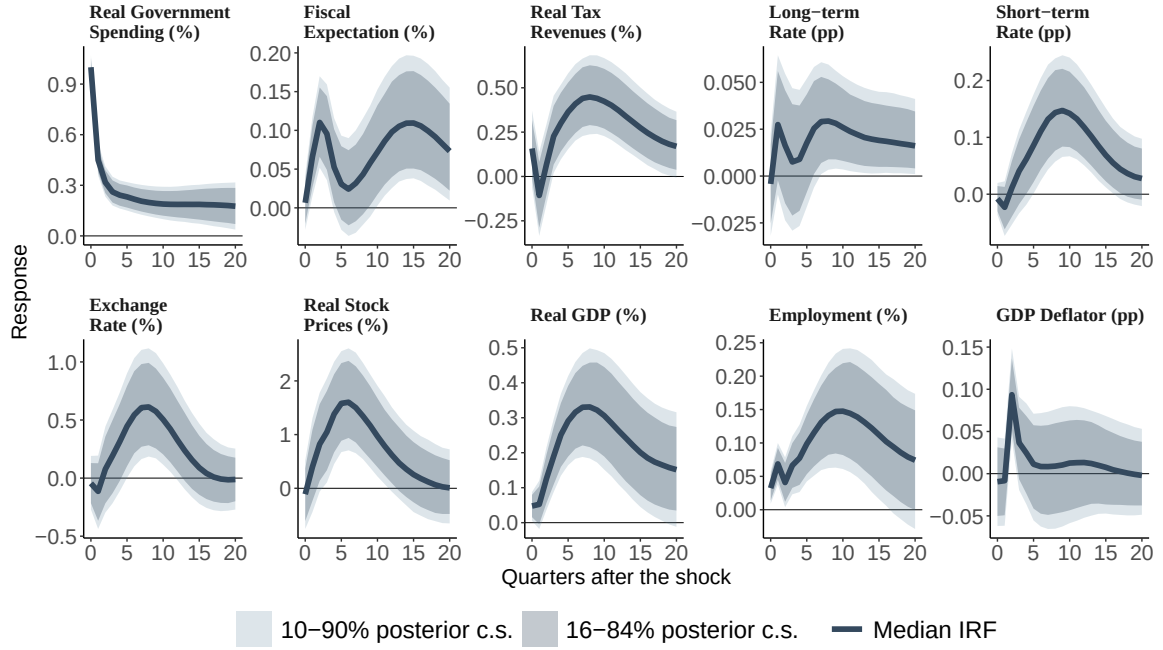
The results section proceeds in three steps. First, we present impulse responses from a benchmark structural VAR (SVAR) estimated on aggregate data, using standard specification and identification methods from the literature. The benchmark model serves as a reference point for the multi-sector analysis. Second, we show that when the sectoral responses implied by the sectoral GVAR model are aggregated, they closely track the responses obtained from the benchmark aggregate SVAR. This comparison provides a validation of the multi-sector framework and ensures consistency with established aggregate evidence. Finally, we turn to the disaggregated results: we present the sectoral responses, the contribution of each sector to the aggregate multiplier highlighting the heterogeneity in the effects of government spending shocks and the role of the production network in shaping these dynamics.

4.1 Benchmark aggregate economy SVAR

We first estimate a Bayesian SVAR using aggregate data as a benchmark for the multi-sector results. The specification mirrors the common-factor block of the GVAR and augments it with aggregate output, employment, and the GDP deflator. The model therefore contains real government purchases, fiscal expectations, real tax revenues, short- and long-term interest rates, the real effective exchange rate, real stock prices, real GDP, employment, and the GDP deflator. The government spending shock is identified using the same recursive scheme as in the GVAR.

Figure 5 shows the impulse responses of 1% unexpected increase in total government purchases. Estimation is bayesian, with a literature standard Normal-Inverse Wishart prior

Figure 5: Benchmark Aggregate economy SVAR



Notes: This figure shows the dynamic impact of a 1% unexpected UK Government purchases on key macroeconomic variables. The lines show the median impulse responses. The grey shades indicate the 68% and 80% central posterior credible set (c.s).

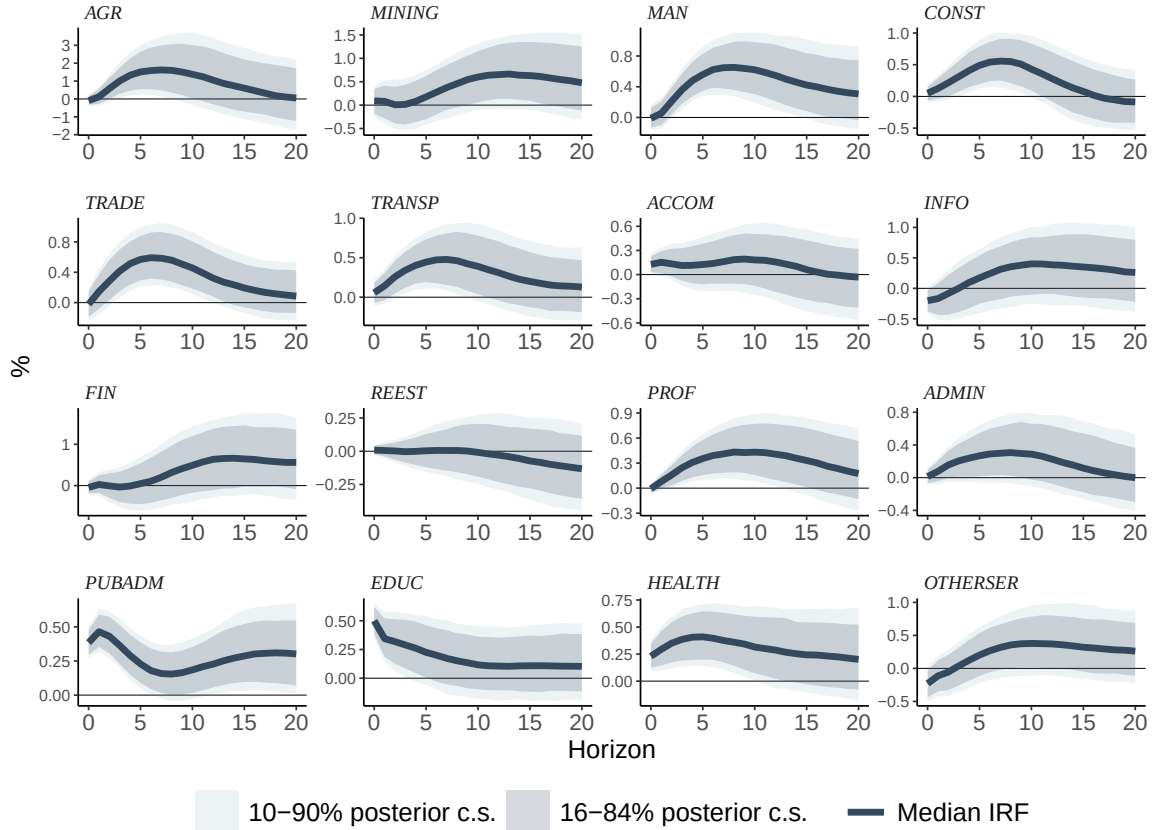
specification. The solid lines show the median IRF and the shaded areas indicate 68% and 80% percent posterior credible set of 5000 draws. The horizontal axis shows quarters after the shock.

Results demonstrate a strong positive response in line with a positive demand shock but without any substantial inflationary pressure. The real government spending reacts strongly to the shock, showing a sharp but short-lived increase. It drops shortly after but remains above the pre-shock level for more than 20 quarters. The fiscal expectations response is positive and in line with the positive and persistent response of the government spending variable. Real GDP rises in a hump-shaped manner, while employment responds with a slight delay and tax revenues increase as economic activity strengthens. Financial variables respond strongly. Both short and long-term interest rates rise, indicating either an endogenous monetary tightening in response to stronger activity or upward pressure from increased government borrowing. Real stock prices rise and the British pound appreciates against foreign currencies consistent with elevated interest rates and capital inflows. Despite these real and financial effects, the response of prices remains muted, with GDP deflator showing only small and short-lived positive deviations from zero.

Overall, the benchmark responses are consistent with a conventional expansionary government spending shock and provide the aggregate reference point against which we evaluate the multi-sector model below.

4.2 Multi-sector GVAR model results

Figure 6: Sectoral GVA impulse response functions



Notes: The plot shows the sectoral GVA responses to an aggregate 1% unexpected government spending shock. The lines show the median impulse responses. The shaded areas indicate the 68% and 80% central posterior credible set (c.s.).

We now turn to the sectoral responses implied by the multi-sector GVAR. Figure 6 reports the responses of real GVA across the 16 industries to an unexpected 1% increase in aggregate government purchases. The responses display substantial heterogeneity in both magnitude and timing.

The sectors with the greatest direct exposure to government demand respond most strongly in the short run. Public Administration & Defence, Education, and Health, which together receive more than 90% of government purchases, exhibit immediate increases in GVA following the shock. Their responses are positive and persistent and peak relatively early, consistent with the direct increase in government demand faced by these industries.

The response outside the main government-financed sectors is more gradual. Among goods-producing industries, Agriculture exhibits the largest response, peaking at approximately 1.9%, while Manufacturing and Construction reach peaks of around 0.6% and 0.5%, respectively. Mining displays a smaller and less precisely estimated response. Importantly, these industries receive only a small fraction of government purchases directly, so their delayed responses indicate that the effects of the aggregate fiscal expansion extend beyond the sectors

in which government demand is concentrated.

A similar pattern emerges among market services. Trade, Transportation, and Professional Services respond positively, with peak responses of around 0.5%. The responses of Accommodation, Finance, and Administrative Services are also positive at some horizons but are estimated less precisely. Real Estate, Information & Communication, and Other Services display small negative responses at some horizons, although the corresponding credible sets generally include zero.

Overall, the sectoral evidence reveals a clear distinction between the immediate and subsequent phases of fiscal transmission. The initial response is concentrated in industries with high direct exposure to government purchases, whereas responses in many market sectors emerge more gradually. This pattern is consistent with the fiscal impulse spreading beyond its initial recipients as the shock propagates through the economy. The contribution of these indirect responses to the aggregate multiplier is examined formally below.

4.3 Aggregate Dynamics in the Multi-Sector Model

Figure 7: Aggregate IRF: SVAR vs. Aggregated GVAR



Notes: This figure shows the impulse response functions of a 1% unexpected UK Government purchases on variables. The figure depicts median impulse responses along with their 68% and 90% central posterior credible set (c.s).

How do results from the multisector model compare to the benchmark SVAR? Because the model is constructed so that sectoral outcomes can be mapped consistently into their aggregate counterparts, we aggregate the sectoral responses of GVA, employment, and prices and compare them with the corresponding responses obtained from the aggregate model.

Figure 7 compares the posterior median responses from the benchmark SVAR with the aggregated sectoral responses from the GVAR. Aggregate GVA and employment responses are constructed using the corresponding sectoral aggregation weights, while the aggregate

price response is obtained using sectoral output shares. The solid line and shaded area report the median response and credible set from the benchmark SVAR, while the dashed line reports the aggregate response implied by the sectoral GVAR.

The two models generate very similar aggregate dynamics. The GVAR closely reproduces the magnitude and timing of the benchmark responses for output, employment, and prices, despite allowing for substantial heterogeneity in the underlying sectoral dynamics.

The close correspondence between the two models provides strong support for the empirical validity of our proposed framework. By matching well-established aggregate dynamics, the model demonstrates that incorporating sectoral heterogeneity and production network linkages does not come at the expense of consistency with aggregate evidence. This validation step ensures that the multi-sector structure is disciplined by the data at the macroeconomic level, thereby providing a credible foundation for the subsequent analysis of sectoral responses and the propagation of fiscal shocks across the economy.

4.4 Decomposing the Aggregate Government Spending Multiplier into Sectoral Contributions

We now exploit the mapping between the sectoral and aggregate output responses to decompose the aggregate government spending multiplier into the contributions of individual sectors. This decomposition requires some care. A conventional sector-specific fiscal multiplier would measure the change in sector i 's output associated with a one-pound increase in government purchases directed specifically towards that sector. This is not the fiscal experiment identified in our model. The aggregate government spending shock raises purchases simultaneously across industries according to their historical expenditure shares, while sectoral output responses also incorporate spillovers transmitted through the production network. Consequently, the response of sector i 's GVA cannot be attributed solely to the government purchases received directly by that sector.

We therefore begin with the conventional aggregate cumulative multiplier. Following [Mountford and Uhlig \(2009\)](#), the multiplier at horizon h is defined as

$$M^{(h)} = \frac{\sum_{j=0}^h \delta^j \Delta Y^{(j)}}{\sum_{j=0}^h \delta^j \Delta G^{(j)}} \times \frac{\bar{Y}}{\bar{G}}, \quad (13)$$

where $\Delta Y^{(j)}$ and $\Delta G^{(j)}$ denote the impulse responses of aggregate output and government purchases at horizon j , respectively, and δ is the discount factor. Since the variables enter the model in logarithms, the first term is the cumulative elasticity of output with respect to government purchases, while the ratio $\bar{Y}/\bar{G}$ converts this elasticity into pound-equivalent units.

Applying the same formula separately to each industry would be misleading for two reasons. First, the sectoral GVA response reflects both the direct effect of government purchases and indirect effects originating elsewhere in the economy. Dividing this response by sector-specific government purchases would therefore attribute network spillovers to spending in sector i alone. Second, several industries receive only a very small share of aggregate government purchases. Scaling their responses by these small expenditure flows would mechanically generate very large and difficult-to-interpret sector-specific multipliers. We instead ask a different question: how much of the aggregate fiscal multiplier is accounted for by the output response of each industry?

Let $s_i = \frac{1}{T} \sum_{t=1}^T s_{i,t}$, where $s_{i,t} = GVA_{i,t}/GDP_t$, denote sector i 's historical average share of aggregate GVA. Since the aggregate output response is constructed from the sectoral responses,

$$\Delta Y^{(j)} = \sum_{i=1}^N s_i \Delta y_i^{(j)}, \quad (14)$$

where $\Delta y_i^{(j)}$ denotes the GVA response of sector i at horizon j . We define sector i 's cumulative contribution to the aggregate output response at horizon h as

$$\theta_i^{(h)} = \frac{\sum_{j=0}^h \delta^j s_i \Delta y_i^{(j)}}{\sum_{j=0}^h \delta^j \Delta Y^{(j)}}. \quad (15)$$

By construction, $\sum_{i=1}^N \theta_i^{(h)} = 1$ so that $\theta_i^{(h)}$ measures the share of the cumulative aggregate output response attributable to sector i . Contributions can be negative when a sector's cumulative output response is negative.

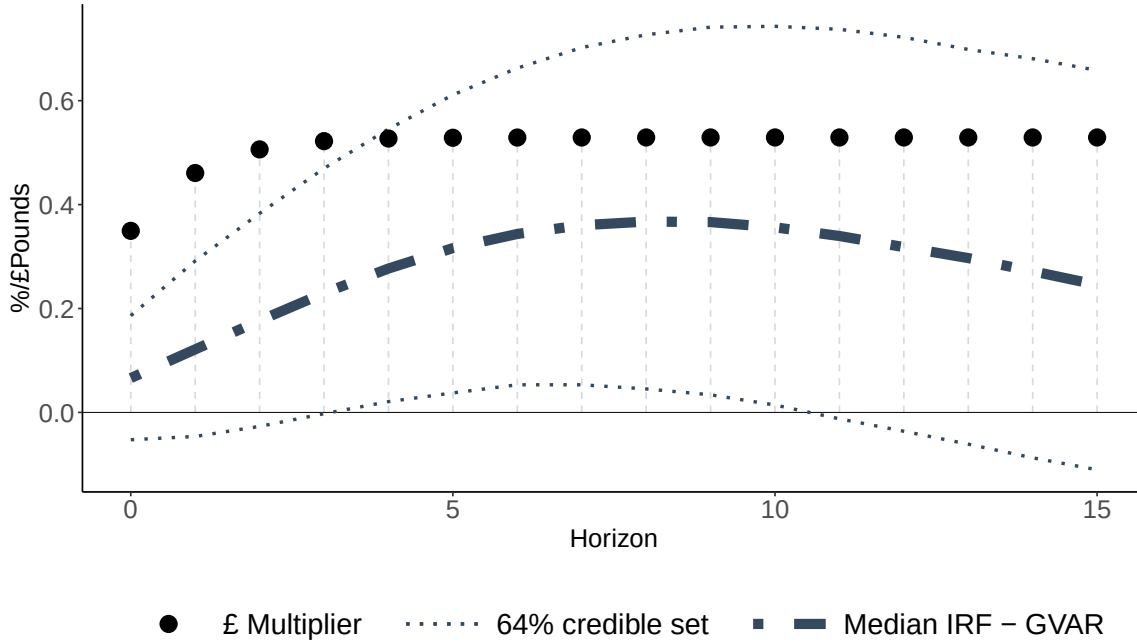
We use these shares to decompose the aggregate multiplier. Sector i 's contribution to the multiplier is defined as

$$M_i^{(h)} = \theta_i^{(h)} M^{(h)}. \quad (16)$$

It follows immediately that $\sum_{i=1}^N M_i^{(h)} = M^{(h)}$. Importantly, $M_i^{(h)}$ should not be interpreted as the causal multiplier associated with directing one additional pound of government spending exclusively towards sector i . Rather, it measures the pound-equivalent contribution of sector i 's GVA response to the multiplier generated by the historically representative aggregate spending shock identified in our model.

Aggregate Multiplier: Figure 8 shows the aggregate output response obtained as the aggregation of individual sectoral responses, and the aggregate multiplier change for four years after the shock. The dashed line is the median response, whereas the dotted lines indicate the 64% posterior credible set. Dots represent the aggregate multiplier. We use

Figure 8: Aggregate output response - Aggregate multiplier



Notes: Notes: The figure reports the aggregate government spending multiplier and the aggregate output response to an unexpected 1% increase in UK government purchases. The aggregate output response is constructed by aggregating the sectoral GVA responses implied by the GVAR. The dashed line shows the posterior median output response and the dotted lines indicate the posterior credible set. The points report the cumulative government spending multiplier at each horizon.

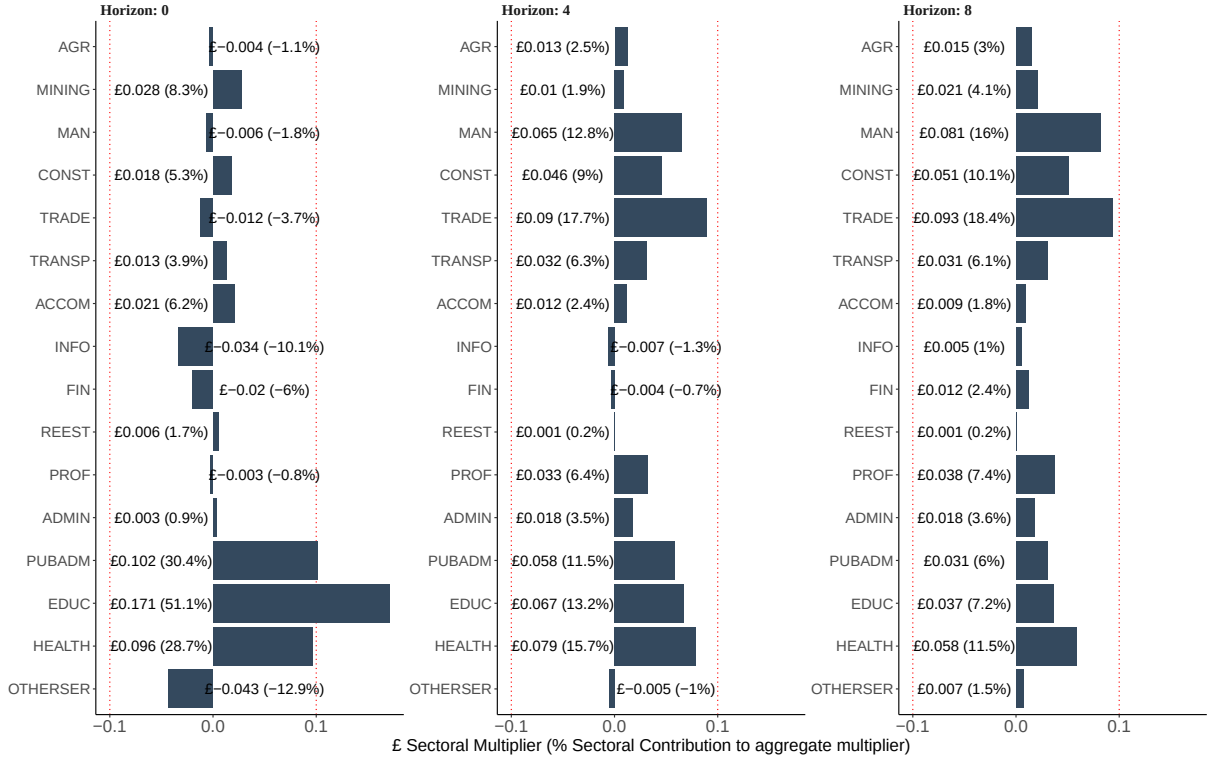
2.5% as a discount factors, which is the 1997-2019 sample average of the short-term interest rate. The sample-average output-to-government-spending ratio, $\bar{Y}/\bar{G}$, is approximately 5.

The multiplier is 0.35 on impact and rises to approximately 0.53 after three quarters, remaining broadly stable thereafter. Although differences in specification and identification prevent a direct comparison, the magnitude of our estimates is broadly consistent with previous estimates of the UK government spending multiplier, including [Glocker et al. \(2019\)](#) and the more recent estimates of [Sousa \(2025\)](#).

Sectoral Contributions: [Figure 9](#) reports each industry’s contribution to the aggregate fiscal multiplier on impact and at horizons of four and eight quarters. By construction, the sectoral contributions sum to the aggregate multiplier at every horizon. Although the aggregate multiplier changes relatively little after the first year, its sectoral composition continues to evolve as the relative contributions $\theta_i^{(h)}$ change over time. The values reported next to each bar give the pound contribution of the sector to the aggregate multiplier, while the percentages in parentheses report its share of the aggregate multiplier.

On impact, the composition of the multiplier closely reflects the direct allocation of government demand. Public Administration & Defence, Education, and Health make the largest contributions, consistent with these sectors receiving more than 90% of government purchases under the historical spending distribution. The immediate aggregate response is therefore

Figure 9: Sectoral Contributions to the Aggregate Government Spending Multiplier



Notes: The figure reports each sector's contribution to the aggregate government spending multiplier following a 1% unexpected increase in UK government purchases. Pound values correspond to $M_i^{(h)} = \theta_i^{(h)} M^{(h)}$, while percentages in parentheses report each sector's share of the aggregate multiplier. These quantities should not be interpreted as multipliers associated with directing one additional pound of government spending exclusively towards a particular sector.

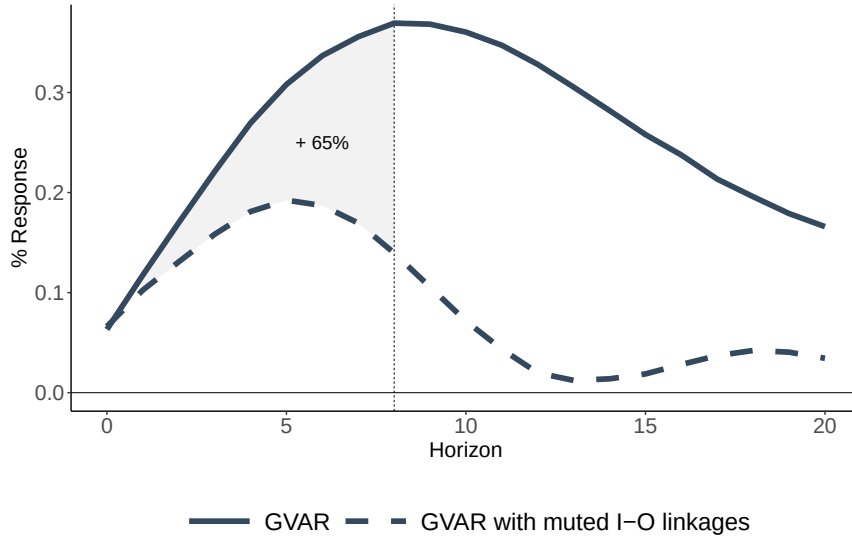
driven predominantly by the industries most directly exposed to the fiscal expansion.

This composition changes markedly as the shock propagates through the economy. At longer horizons, the contribution of the three initially dominant sectors declines, while the contribution of market industries increases. One and two years after the shock, Trade, Manufacturing, Construction, and Health are among the largest contributors, accounting for approximately 18%, 16%, 10%, and 10% of the aggregate multiplier, respectively. Together, these sectors account for more than half of the aggregate response. Thus, even though the aggregate multiplier stabilises relatively quickly, the industries underlying it continue to change as the fiscal impulse spreads beyond its initial recipients.

4.5 Assessing Network Effects: A Counterfactual Analysis

The aggregate multiplier is the outcome of a simultaneous and uneven increase in government spending across sectors. The dynamics of the sectoral responses suggest that the production network plays a central role in the propagation of the shock. In the short run, the impact is concentrated in the subset of sectors receiving the largest share of government purchases. However, in later horizons the shock propagates to the rest of the economy through

Figure 10: The role of the production network in aggregate output response



Notes: This figure shows the impulse response functions of a 1% unexpected UK Government purchases on aggregate output. The solid line represents the aggregated sectoral output responses of the baseline model. The dashed line represents the aggregated sectoral output responses with muted cross-sectoral interactions.

input–output linkages, affecting a broader set of industries. But, how much does the production network determine the aggregate fiscal multiplier? To assess the importance of these network effects, we conduct a counterfactual exercise that quantifies how the production structure amplifies or attenuates the impact of the fiscal shock.⁹

We quantify the role of network effects by comparing the responses from the baseline GVAR model with a set of responses where the cross-sectoral channel coefficients are muted. The difference between the baseline and counterfactual responses allows us to disentangle the direct effects operating through the common factor block from the indirect effects, arising from inter-sectoral trade. To calculate the counterfactual responses, we partition matrix G_0 so that contemporaneous correlations among sectors are dropped. We then calculate sectoral responses as in the baseline case.

Figure 10 contrasts the two responses. The solid line represents the aggregated sectoral output responses of the baseline model and the dashed line represents the counterfactual response. The grey areas indicate the 8-quarter ahead cumulative effects. The 8-quarter multiplier from the counterfactual output response is 0.19. That is that sectoral linkages account for about $(1 - \frac{0.19}{0.55}) \times 100 = 65\%$ of the overall government purchases multiplier.

The shape of the simplified IRF aligns with the concept of gradual propagation of shocks through the production network. On impact the IRFs move closely, capturing the direct from reflecting the simultaneous increase of spending across industries. The distance between the two lines gets pronounced over the cross-sectoral channel is diffused and the higher order effects are not captured.

⁹Similar approach is used in Dragomirescu-Gaina and Elia (2021) and Burriel and Galesi (2018).

However, the importance of the sectoral linkages might be underestimated here as the effect of common variables (that is not shut down) may capture some of the common factor effects on sectoral variables.

5 Conclusion

This paper develops an empirical framework for uncovering the sectoral origins of the aggregate government spending multiplier. We combine aggregate macroeconomic data, sectoral time-series, and Input–Output tables within a Bayesian GVAR that allows individual industries to exhibit heterogeneous dynamics while interacting through the production network. The fiscal experiment corresponds to an unexpected increase in aggregate government purchases that preserves their historically representative allocation across industries. This structure provides a direct mapping from sectoral output responses to the aggregate response and allows the government spending multiplier to be decomposed into the contributions of individual sectors.

The results show that substantial heterogeneity lies beneath the aggregate fiscal response. The multi-sector model closely reproduces the aggregate dynamics obtained from a conventional SVAR, while revealing large differences in the magnitude and timing of sectoral responses. The aggregate multiplier is approximately 0.35 on impact and rises to around 0.55 after one year. Initially, the response is concentrated in Public Administration & Defence, Education, and Health, which together receive more than 90% of government purchases. As the shock propagates, however, the composition of the multiplier changes markedly. Trade, Manufacturing, Construction, and Health become among the largest contributors at longer horizons, despite several of these industries receiving relatively little government demand directly. Thus, even when the aggregate multiplier has largely stabilised, the industries underlying that response continue to change.

Production linkages are central to this propagation. Counterfactual exercises that mute intersectoral Input–Output linkages substantially reduce the aggregate multiplier relative to the baseline, indicating that a sizeable share of the fiscal response is generated through cross-sectoral transmission. The aggregate effects of government spending therefore depend not only on the industries directly receiving government demand, but also on how the initial fiscal impulse propagates through the production network. This distinction helps explain why sectors with limited direct exposure to government purchases can nevertheless make sizeable contributions to the aggregate response at longer horizons.

More broadly, the analysis highlights the information that is lost when fiscal policy is studied exclusively through aggregate variables. An aggregate multiplier summarises the overall output effect of government spending, but conceals both the industries responsible for that response and the intersectoral interactions through which it develops over time. The decomposition proposed in this paper provides a way of recovering this underlying structure from the data. At the same time, the sectoral contributions estimated here should not be

interpreted as causal multipliers from directing an additional pound of government spending exclusively towards a particular industry. They describe the sectoral composition of the response to a historically representative aggregate fiscal expansion. Extending the framework to alternative expenditure compositions, different production-network structures, and other economies provides a natural direction for future research.

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A Additional Tables

Table A1: Mapping of SIC sectors to ISIC Rev. 4

Abbreviation	SIC Code	ISIC	ISIC Description
AGR	A	A	Agriculture, Forestry and Fishing
MINING	BDE	B, C, D	Mining and Quarrying (B); Manufacturing (C); Electricity, Gas, Steam and Air Conditioning Supply (D)
MAN	C	C	Manufacturing
CONST	F	F	Construction
TRADE	G	G	Wholesale and Retail Trade; Repair of Motor Vehicles and Motorcycles
TRANSP	H	H	Transportation and Storage
ACCOM	I	I	Accommodation and Food Service Activities
INFO	J	J	Information and Communication
FIN	K	K	Financial and Insurance Activities
REEST	L	L	Real Estate Activities
PROF	M	M	Professional, Scientific and Technical Activities
ADMIN	N	N	Administrative and Support Service Activities
PUBADM	O	O	Public Administration and Defence; Compulsory Social Security
EDUC	P	P	Education
HEALTH	Q	Q	Human Health and Social Work Activities
OTHERSER	RST	R, S, T	Arts, Entertainment and Recreation (R); Other Service Activities (S); Activities of Households as Employers; Undifferentiated Goods- and Services-Producing Activities of Households for Own Use (T)

Note: The table maps the 16 sector disaggregation SIC codes used in this study to their corresponding ISIC Rev. 4 codes and descriptions. Sectoral aggregation corresponds to a one-digit industry classification with some sectors further aggregated to ensure consistency with available quarterly time-series data.

B Descriptive tables and figures

Table B2: Data definitions and sources

Variable in VAR	Definition	Construction	Source
Panel A: Total economy data			
G_t	Government spending	log-level	OECD
ΔG_t^{fc}	Government spending growth forecast	growth rate	OECD
T_t	Tax revenues	log-level	OECD
R_t^S	Short term interest rate	rate	OECD
R_t^L	Long term interest rate	rate	OECD
S_t	Real exchange rate (USD per GBP)	log-level	OECD
V_t	Equity market	log-level	OECD
Panel B: Sectoral data			
Y_t	Sectoral GVA	log-level	ONS
N_t	Employment	log-level	ONS
P_t	Sectoral GVA deflator	log-level	ONS

Notes: Data span the period 1997Q1–2019Q4. IO tables are published in annual frequency are available for the same period. Sectoral data and IO tables can be found here: [-OECD Input-Output Tables-](#), [-Sectoral GVA-](#), [-Sectoral employment-](#).

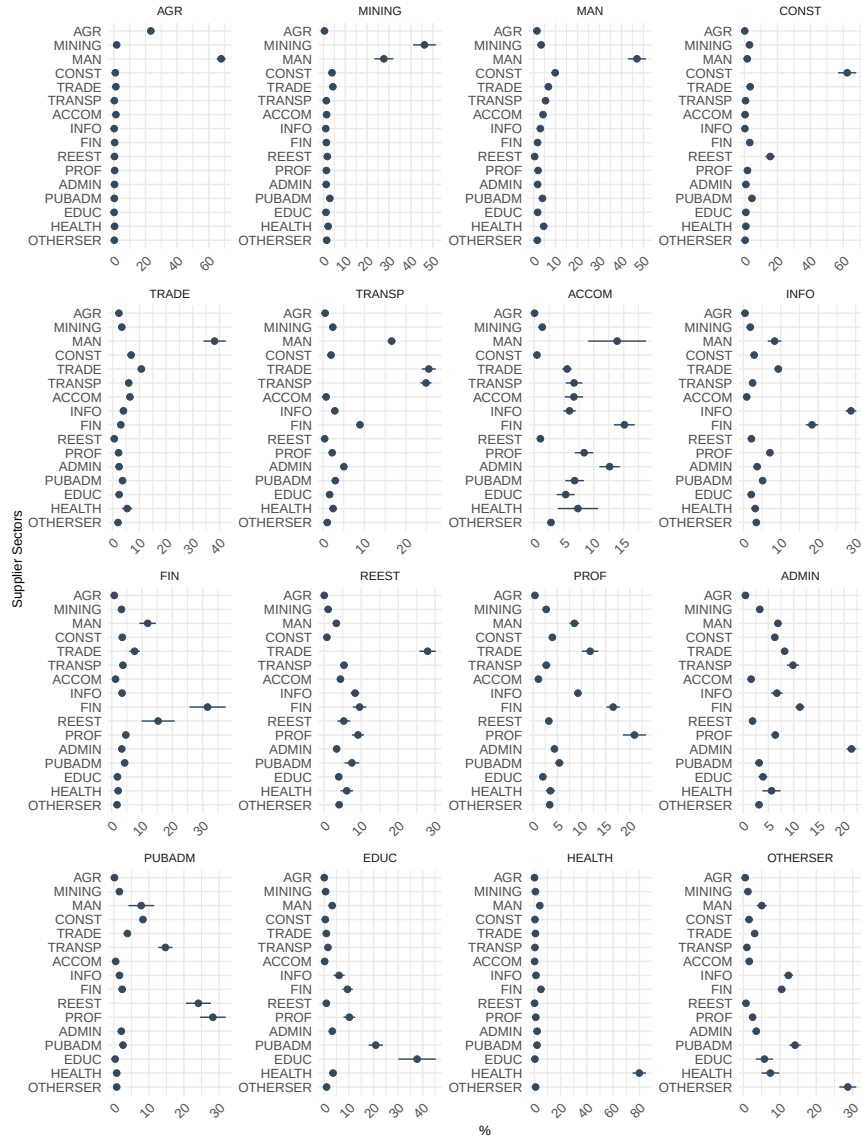
We proxy fiscal expectation with government spending projections from historical vintages of the OECD Economic Outlook (EO). As EO projections are published biannually, we interpolate the fiscal expectations series to a quarterly frequency using midpoint averaging. Although the EO data are of lower frequency, OECD historical forecasts remain particularly useful due to their long temporal coverage. Alternative sources of historical forecasts for the UK economy, such as those produced by the Office for Budget Responsibility, span only a limited period and were therefore not preferred.

Table B3: Sectoral allocation of GFCE (percentage of total GFCE)

Sector	Mean (%)	SD (%)	Max (%)	Min (%)
Agriculture (AGR)	0.008	0.00	0.01	0.00
Mining (MINING)	1.82	0.12	1.96	1.54
Manufacturing (MAN)	1.46	0.13	1.65	1.19
Construction (CONST)	0.00	0.00	0.00	0.00
Trade (TRADE)	0.77	0.07	0.89	0.63
Transport (TRANSP)	0.52	0.15	0.84	0.25
Accommodation & Food (ACCOM)	0.00	0.00	0.00	0.00
Information & Communication (INFO)	1.00	0.22	1.41	0.69
Financial Services (FIN)	0.00	0.00	0.00	0.01
Real Estate (REEST)	0.00	0.00	0.00	0.00
Professional Services (PROF)	0.12	0.01	0.14	0.09
Administrative Services (ADMIN)	0.01	0.00	0.02	0.00
Public Administration & Defence (PUBADM)	38.2	2.89	43.0	34.6
Education (EDUC)	16.6	0.81	17.5	14.8
Health (HEALTH)	38.2	3.89	44.2	32.5
Other Services (OTHERSER)	1.27	0.29	1.65	0.80

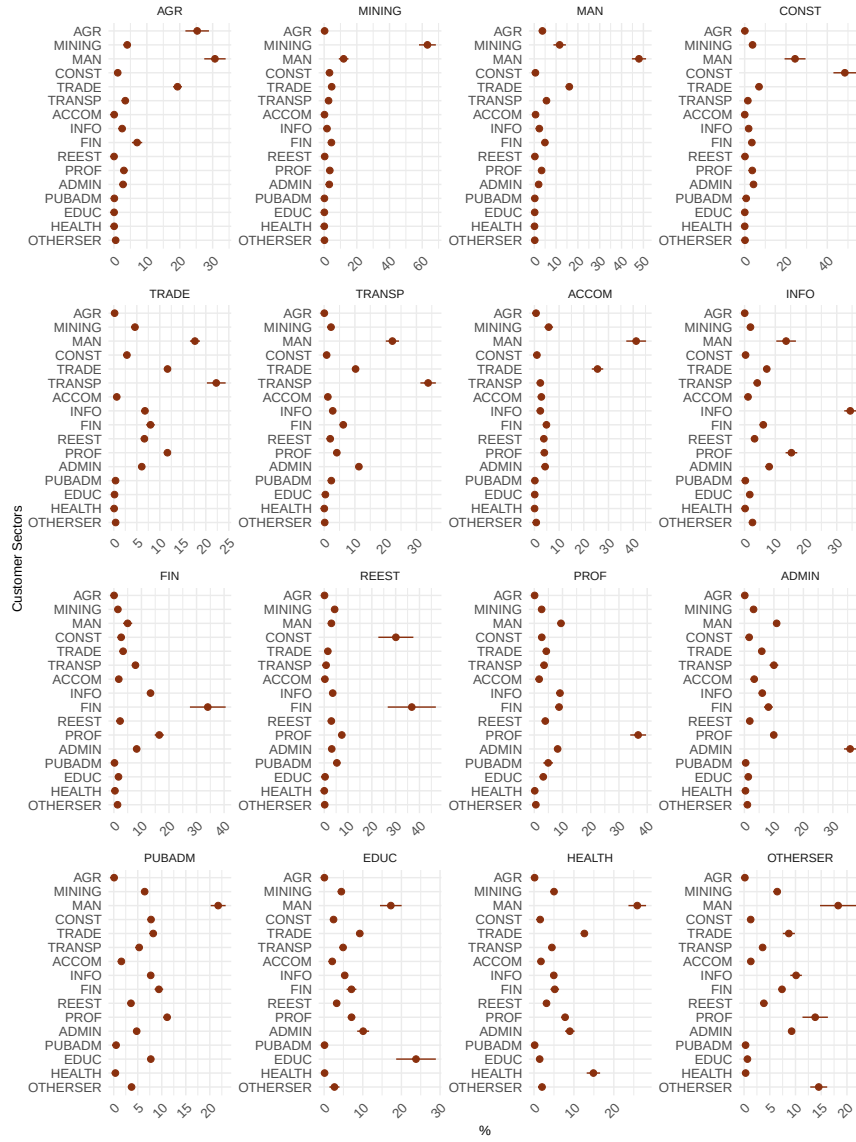
Note: The table reports the mean, standard deviation, maximum, and minimum share of government spending across sectors as a percentage of total GFCE for the 1997–2019 period.

Figure B1: Customer-Supplier IO matrix



Notes: The plot shows the distribution of the row-normalised Supplier-Customer IO matrix over the 1997-2019 period. Points indicate mean values while segments show one standard deviation.

Figure B2: Customer-Supplier IO matrix



Notes: The plot shows the distribution of the row-normalised Customer-Supplier IO matrix over the 1997-2019 period. Points indicate mean values while segments show one standard deviation.